

# AIMS DTU Research Intern 2026

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## LLM Jailbreaking and Safety

### Preview

As large language models become increasingly integrated into real-world systems, ensuring their reliability and safety has become a critical challenge. Despite extensive alignment and moderation training, these models can still be manipulated through adversarial prompting techniques that bypass intended safeguards. This phenomenon, commonly referred to as LLM jailbreaking, exposes weaknesses in how models interpret instructions, prioritize context, and enforce safety constraints.

A Crescendo attack is a sophisticated, multi-turn AI jailbreaking technique where an adversary bypasses an LLM's safety guardrails through gradual conversational escalation rather than a single malicious prompt. Unlike single-prompt jailbreaks, Crescendo attacks exploit:

1. Memory stacking
2. Guard-lowering dialogue
3. Semantic drift
4. Prompt disguising

### Your Assignment

Devise a defense pipeline around Llama-3.2-3B-Instruct that intercepts and neutralizes the dialogue-based escalation (Crescendo technique).

The task consists of three major stages:

- Build a defensive framework around Llama-3.2-3B-Instruct to identify, prevent, and mitigate Crescendo-style multi-turn jailbreak attacks.
- Implement at least two distinct mitigation approaches that address conversational escalation techniques.
- Benchmark the proposed defenses against 10 simulated Crescendo attack vectors and compare their effectiveness using metrics such as Attack Success Rate (ASR).

You may use datasets such as the [Argilla DPO Mix 7K](#), or any other publicly available conversational corpora, synthetic data generation methods, or custom-curated datasets depending on the proposed defense approach. The choice of resources and the reasoning behind them should be documented in the final report.

## **Evaluation**

Evaluate the effectiveness of your defense mechanisms using:

- Attack Success Rate (ASR) as the primary metric for measuring defense effectiveness
- Comparative analysis among your different defense strategies
- Effectiveness against diverse Crescendo attack vectors
- Robustness of each mitigation approach
- Resource efficiency and computational overhead of the defense pipeline

## **Deliverables**

1. A technical report (4-6 pages) covering your experimental setup, results, comparative analysis among different strategies, limitations, and directions for future work.
2. A GitHub repository with clean, reproducible code for the pipeline (fine-tuning, processing, or any other devised strategy). Model weights may be uploaded to Google Drive with a link included in the report.